

Report

Title: Experimenting with Volumes on Windows VM

Setup

Creating a fresh EC2 instance with following configs;

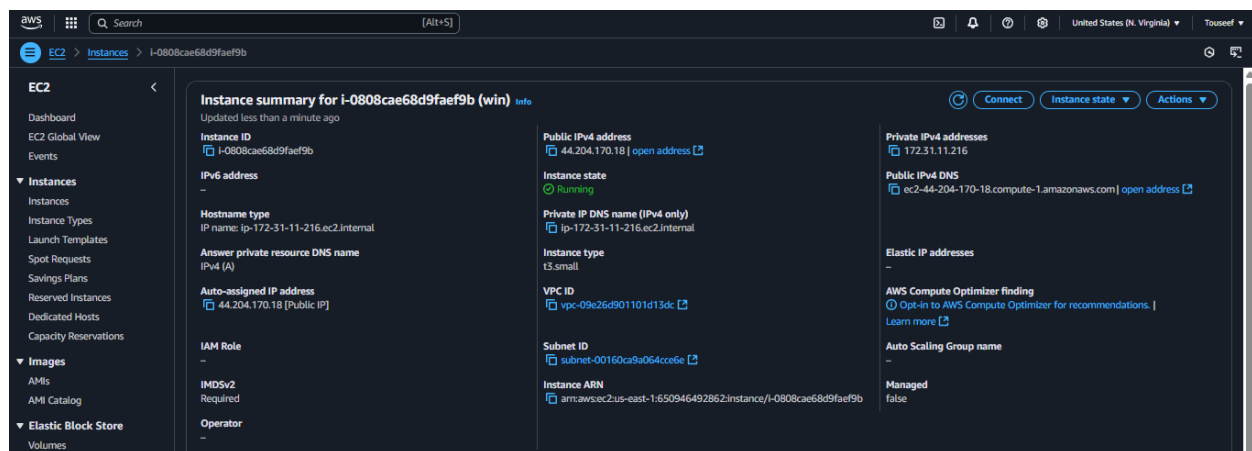
- AMI => Windows OS
- Instance Type => T3.micro
- Create New Key Value Pair Encryption (it will download the .pem file on your machine)
- Select a zone from subnet inside Network Details
- Allow RDP traffic
- Keep other at Default

Procedure

Create a fresh windows VM using the setup described above in the zone us-east-1a by setting the subnet as shown below;

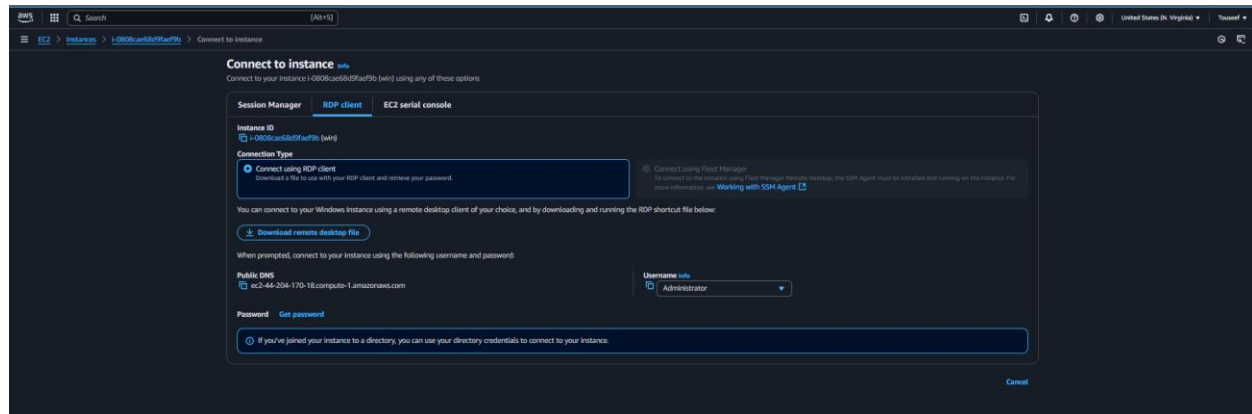


Go to Connect section of your instance.

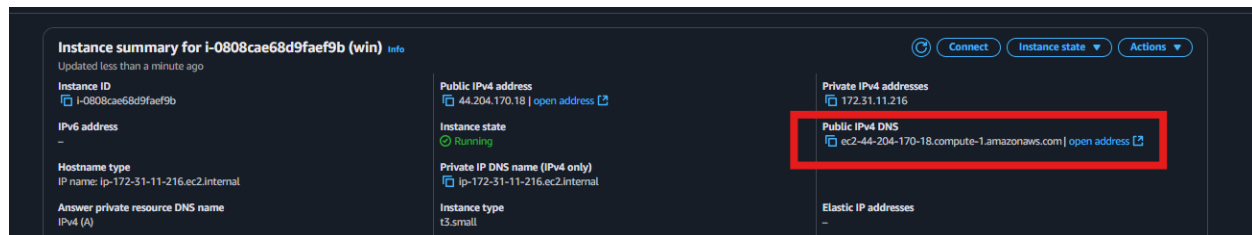


Go to RDP tab, click on get password link, upload your .pem file (which was downloaded when create key pairs) to decrypt the password.

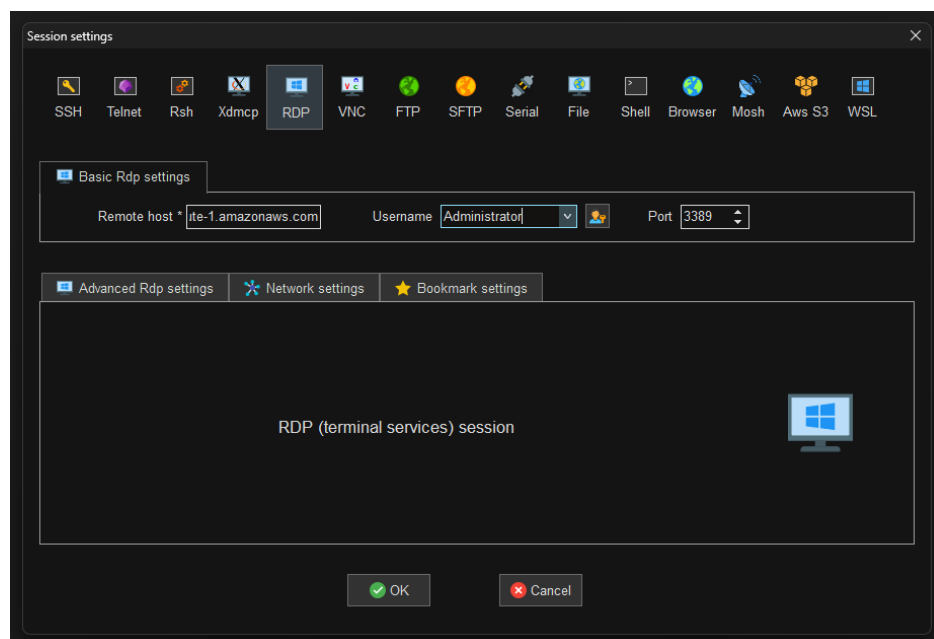
Note down the username and password to connect to your windows instance in future.

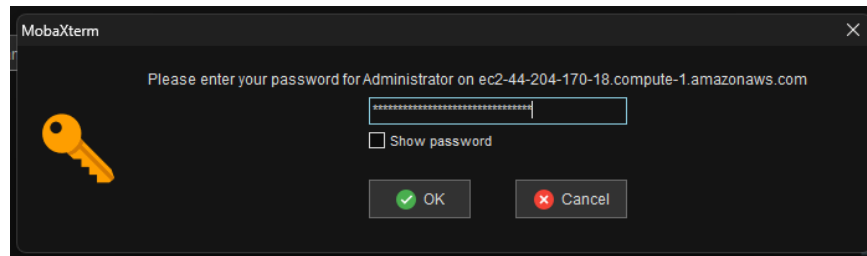


Copy public IPv4 DNS address of your instance as follows;

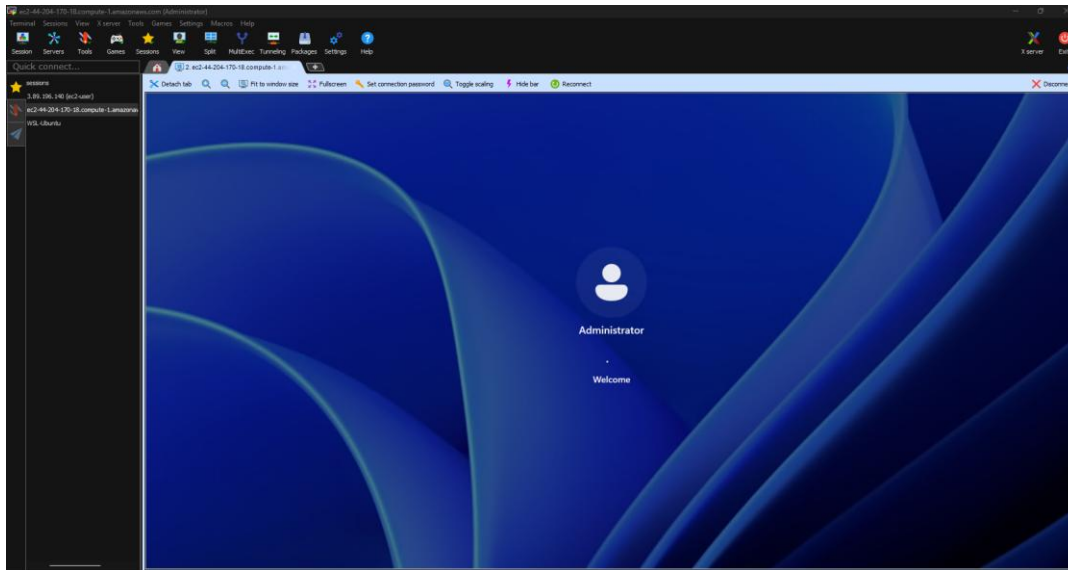


Inside MobaXterm, select Session from the toolbar. Select RDP from the top session types.





Now you will be connected to the instance as follows;



Create a new 1GiB volume in the same zone as the instance as follows;

Create volume Info

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Volume type Info
General Purpose SSD (gp3)

Size (GiB) Info
1
Min: 1 GiB, Max: 16384 GiB.

IOPS Info
3000
Min: 3000 IOPS, Max: 16000 IOPS.

Throughput (MiB/s) Info
125
Min: 125 MiB, Max: 1000 MiB. Baseline: 125 MiB/s.

Availability Zone Info
us-east-1a

Snapshot ID - optional Info
Don't create volume from a snapshot

Encryption Info
Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.
☐ Encrypt this volume

Tags - optional Info

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add tag](#)
You can add 50 more tags.

Snapshot summary Info

[Click refresh to view backup information](#)
The volume type that you select and the tags that you assign determine whether the volume will be backed up by any Data Lifecycle Manager policies.

[Cancel](#) [Create volume](#)

Attach the newly created volume to the newly created windows VM as follows;

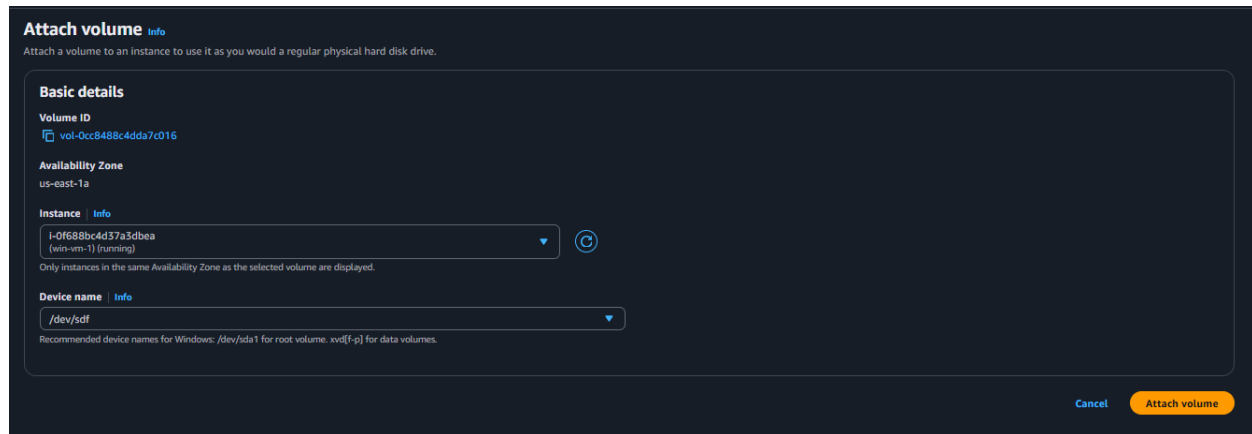
Volumes (1/2) Info

Save filter sets
Choose filter set

	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Created	Availability Zone	Volume state	Alarm status	Attach
☒	-	vol-06262cae04a9d1c45	gp3	1 GiB	3000	125	sn-029c70a6c7	30/11/16 14:18 GMT+5	us-east-1b	Available	No alarms	+
☐	-	vol-08930e9082302b7d5	gp3	30 GiB	3000	125		33/16 14:17 GMT+5	us-east-1b	In-use	No alarms	+

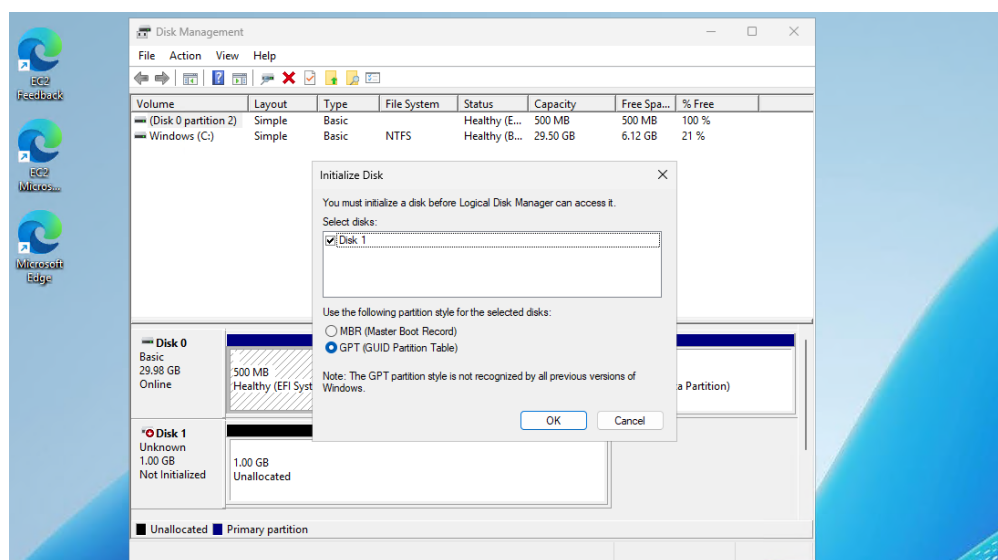
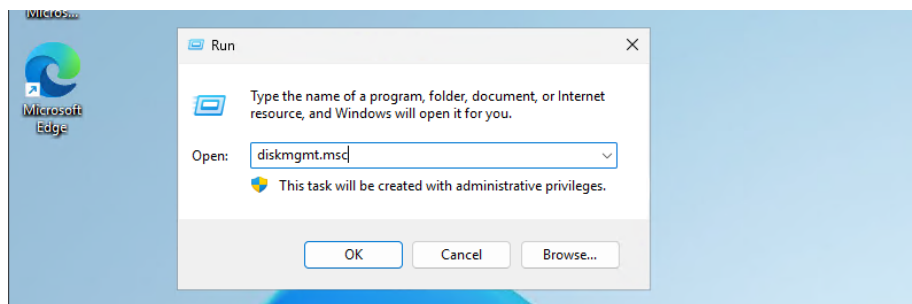
Context menu for selected volume:

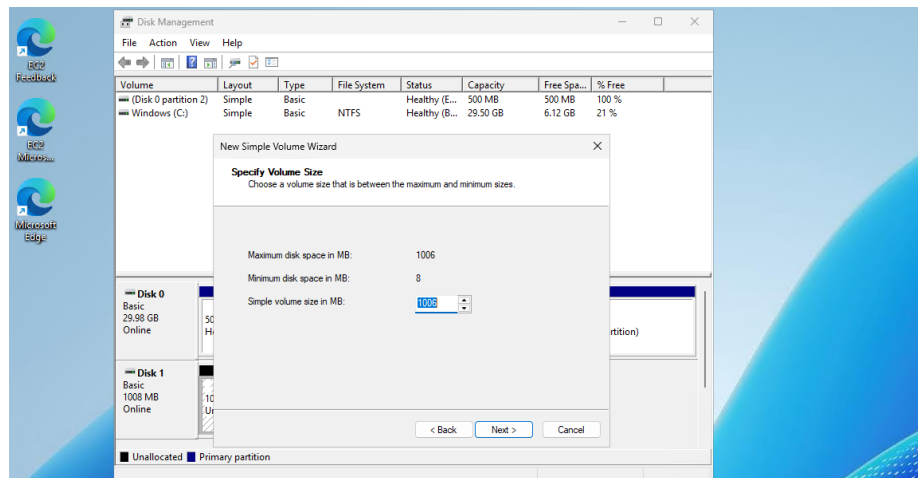
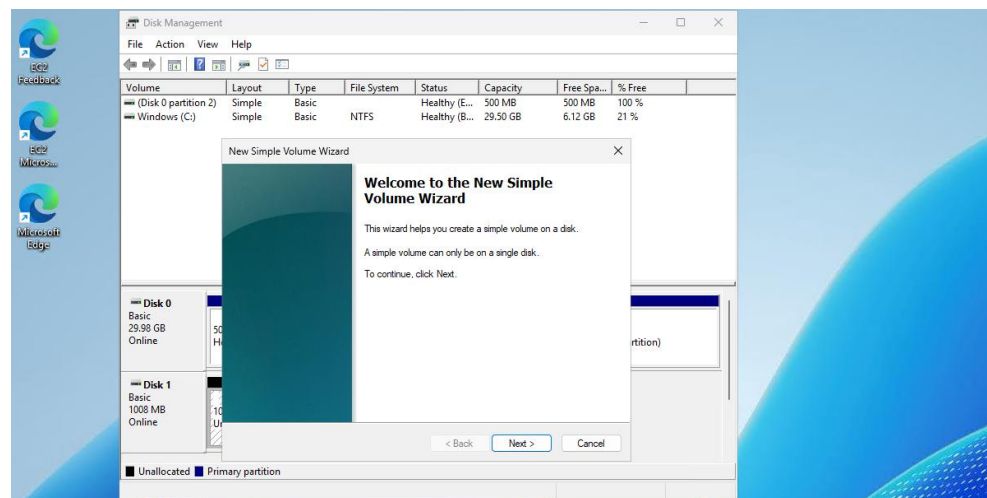
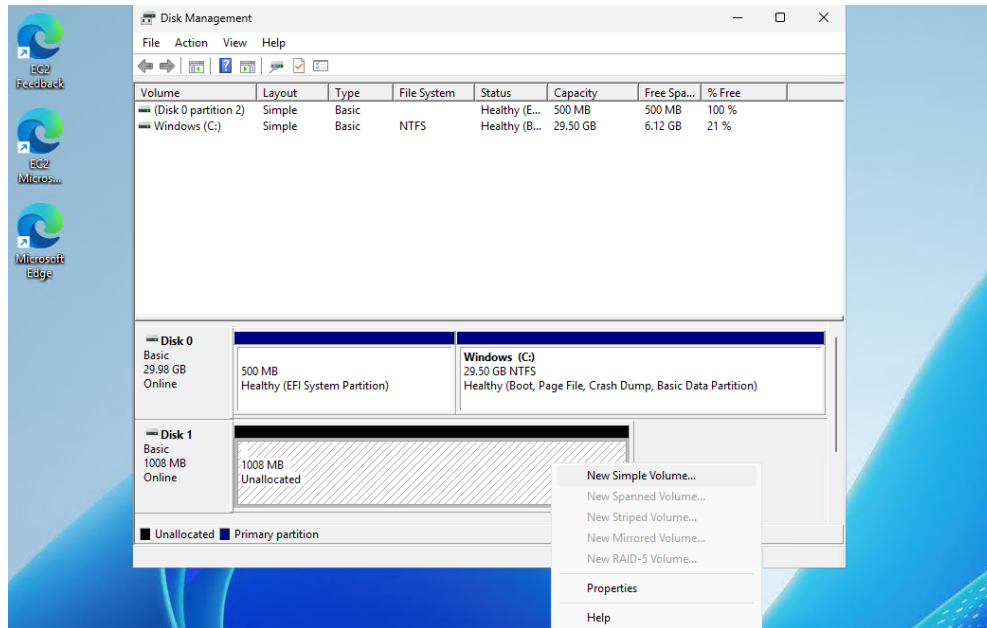
- Create volume
- Modify volume
- Create snapshot
- Create snapshot lifecycle policy
- Delete volume
- Attach volume**
- Detach volume
- Force detach volume
- Manage auto-enabled I/O
- Manage tags
- Fault injection

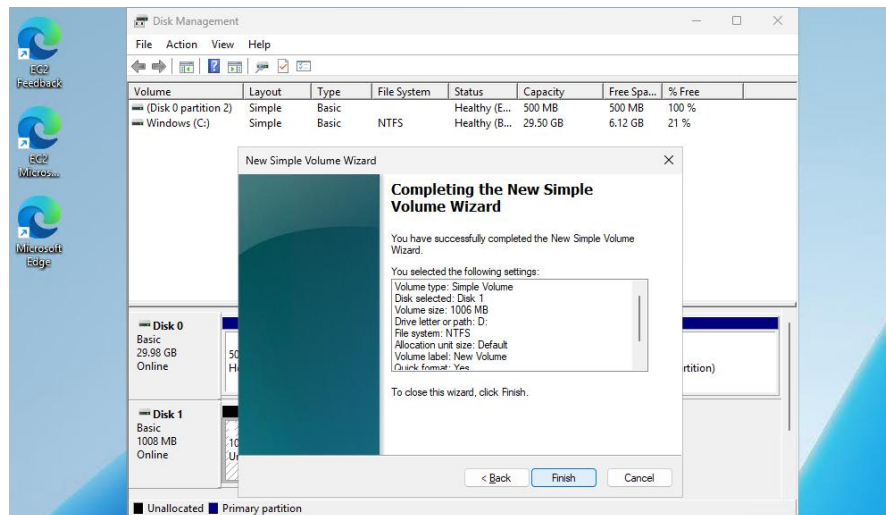
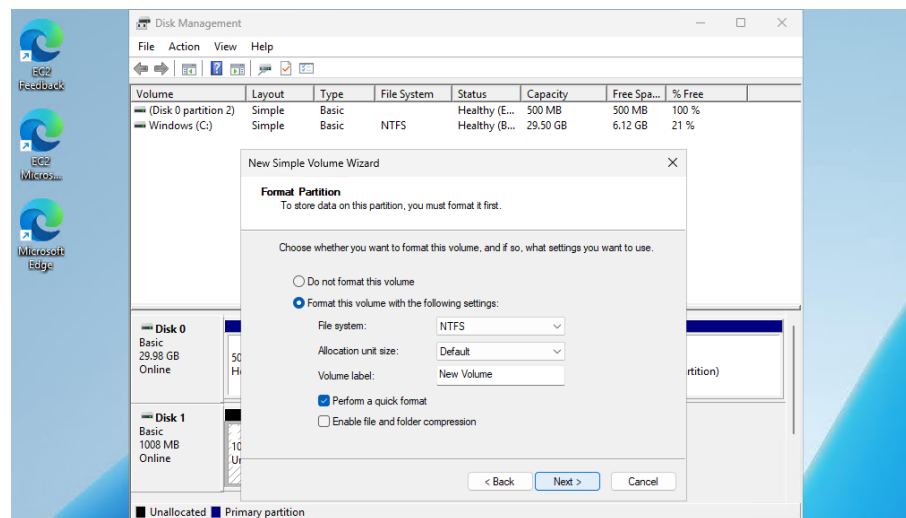
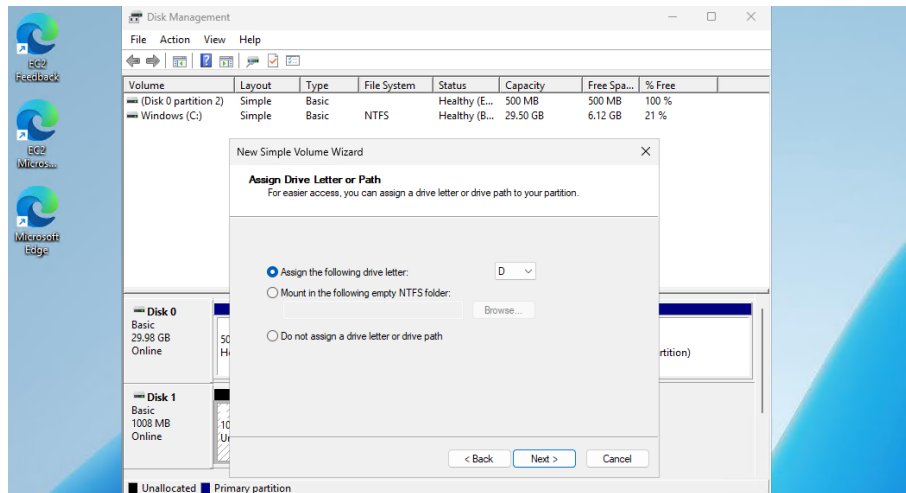


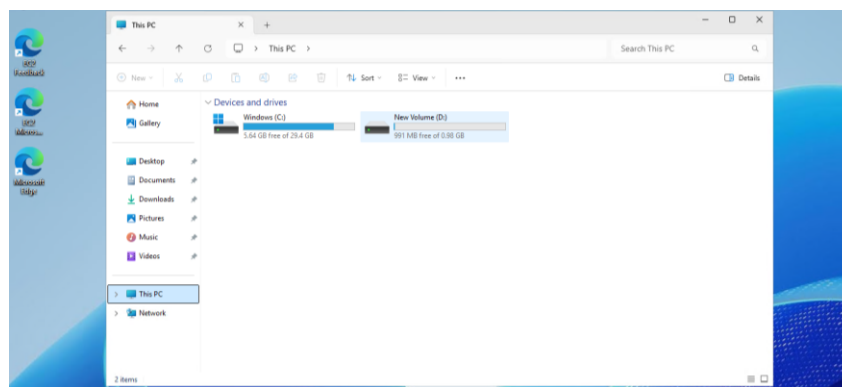
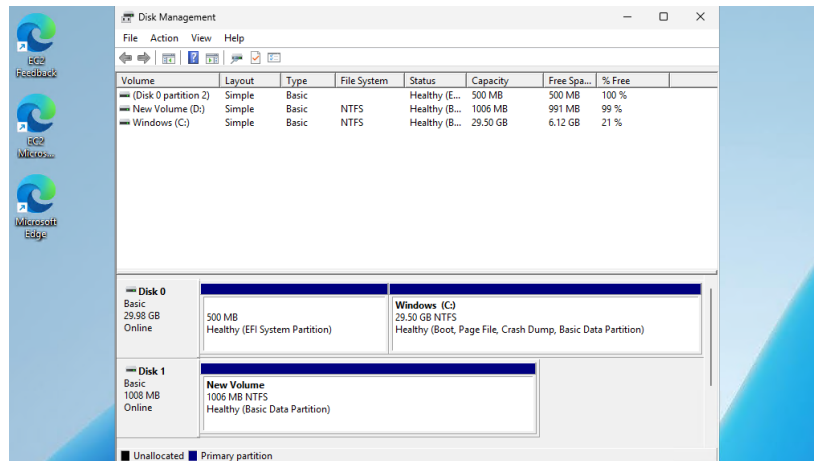
If you are unable to view your instance in the list, then it probably means that the volume and the instance both are in different zones.

Now go to your instance and open the Disk Management Setup of the windows by entering the diskmgmt.msc command in the Run dialogue box as shown in the pictures below.

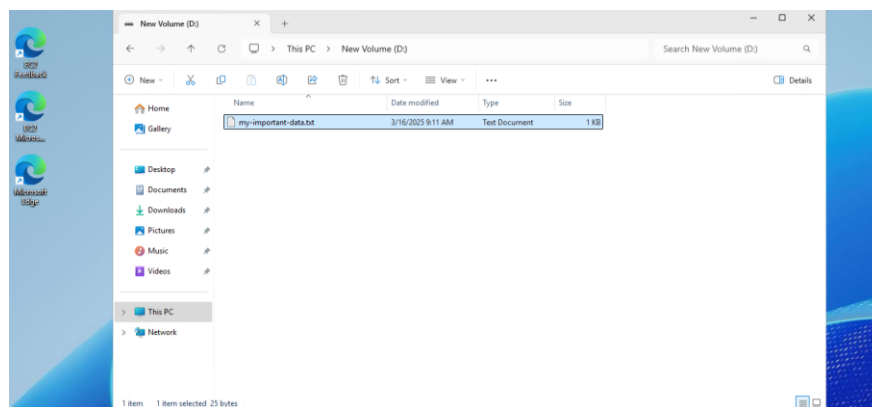




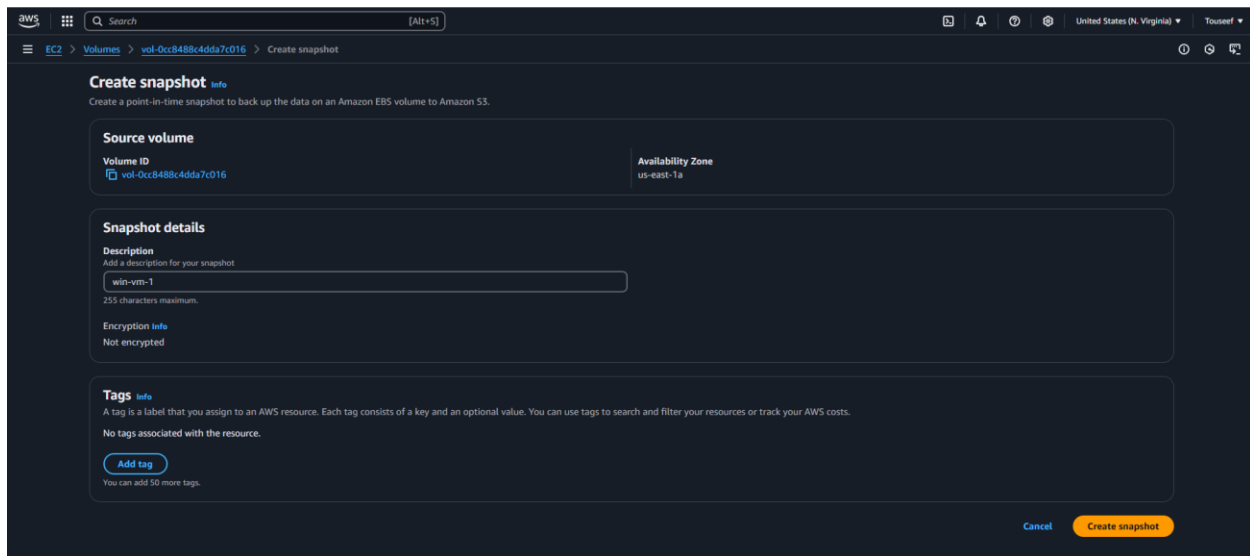
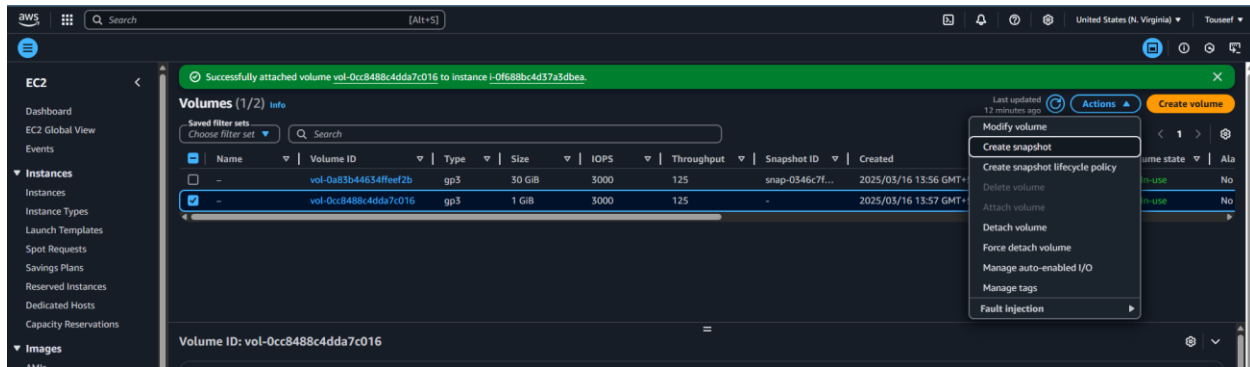




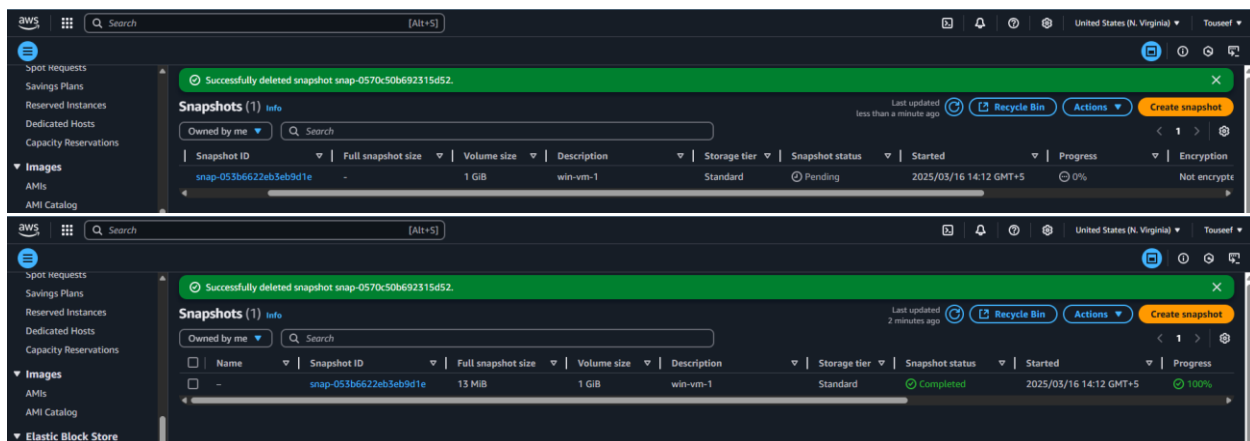
Create a file in the newly mounted volume as shown in the figure below;



Now create a snapshot of the volume as follows;



The snapshot will take time to finish. Make sure it's finished from the progress and status of the snapshot as shown in the pictures below.



Now create a fresh windows VM using the setup described above in the zone us-east-1b by setting the subnet as shown below;

Network settings Info

VPC - required Info

vpc-09e26d901101d13dc (default) ↕

172.31.0.0/16

Subnet Info

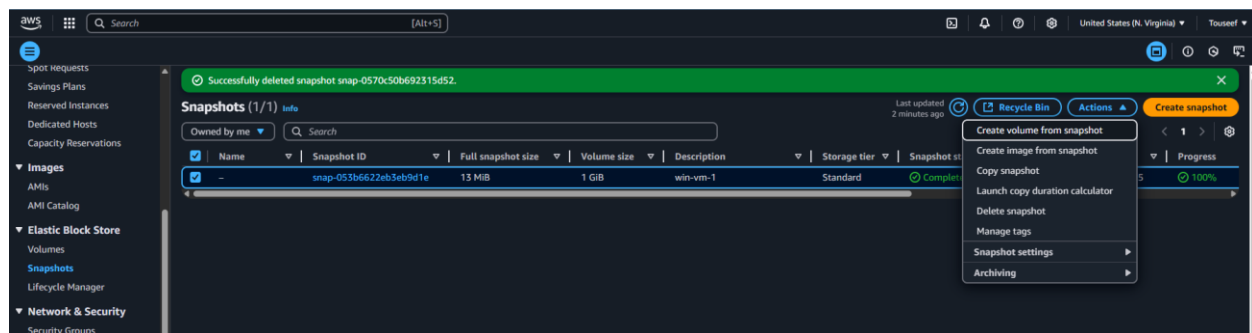
subnet-0168d084f7ea88210

VPC: vpc-09e26d901101d13dc Owner: 650946492862 Availability Zone: us-east-1b

Zone type: Availability Zone IP addresses available: 4091 CIDR: 172.31.16.0/20

↕ Create new subnet ↗

Create a volume from the snapshot (just taken) in the same zone as your instance. Refer to below diagrams for guidance.



Create volume Info

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Snapshot ID

snap-053b6622eb3eb9d1e

Volume type Info

General Purpose SSD (gp3)

Size (GiB) Info

1

Min: 1 GiB, Max: 16384 GiB.

IOPS Info

3000

Min: 3000 IOPS, Max: 16000 IOPS.

Throughput (MiB/s) Info

125

Min: 125 MiB, Max: 1000 MiB. Baseline: 125 MiB/s.

Availability Zone Info

us-east-1b

Fast snapshot restore Info

Not enabled for selected snapshot

Encryption

Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.

☐ Encrypt this volume

Tags - optional Info

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add tag

You can add 50 more tags.

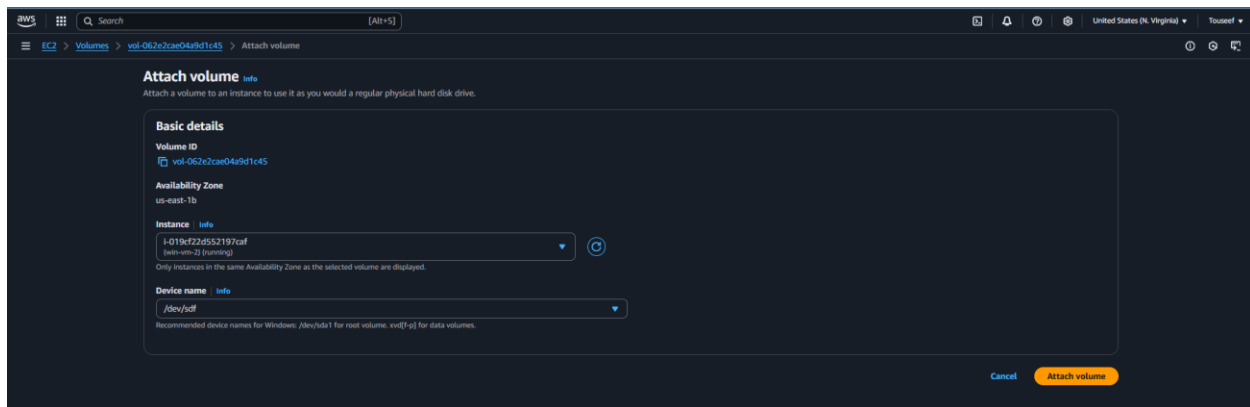
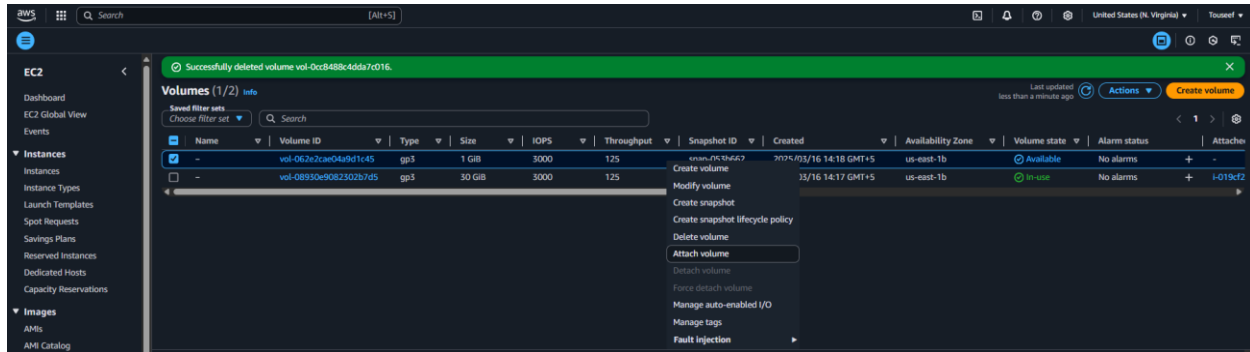
Snapshot summary

Click refresh to view backup information

The volume type that you select and the tags that you assign determine whether the volume will be backed up by any Data Lifecycle Manager policies.

Cancel Create volume

Now attach the newly created volume to newly created windows VM. Refer to the diagram below for guidance.



Now connect to your Windows VM using RDP client from your local machine and open the Disk Management Wizard as we did previously.

Now you can see that the new volume is already available and mounted on drive D. And inside the File Explorer, you will be able to see the same data.

